

Vincent SASSARD *Ph. D. Candidate*

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Research Interests

- Lithosphere-asthenosphere interactions using seismic methods and numerical modeling.
- Subduction zone structure and dynamics using seismic imaging.

Education

2027 West Lafayette, IN	Doctor of Philosophy, Purdue University Advisor: Dr. Xiaotao Yang. Ph. D. Dissertation title: <i>Seismic imaging of the Alaskan lithospheric system from the ocean to the continental interior and its tectonic implications.</i>
2022 Utrecht, The Netherlands	Master's of Science, Universiteit Utrecht Geophysics, Earth Structure & Dynamics. Advisor: Dr. Ylona van Dinther. MSc thesis title: <i>What Controls Lateral Segmentation in Cascadia? Sub-slab Density Anomalies or Slab Age Variations?</i>
01/2020 – 06/2020 Eugene, OR, USA	Semester Abroad, University of Oregon
2020 Poitiers, France	Bachelor's of Science, Université de Poitiers Earth Sciences

Publications

07/2025	Slab tearing along a subducted oceanic plate joint beneath the Alaska Peninsula, <i>Nature Geoscience</i> Sassard, V., Yang, X., Liu, L. <i>et al.</i> Slab tearing along a subducted oceanic plate joint beneath the Alaska Peninsula. <i>Nat. Geosci.</i> 18 , 801–807 (2025). https://doi.org/10.1038/s41561-025-01749-6 
01/2023	SCENTAR: A High-Density Nodal Array to Study the Structure and Seismogenic Behavior of the Southern Cascadia Forearc, <i>Seismological Research Letters</i> Delph, J. R., Thomas, A. M., Stanciu, A. C., Aslam, K., Chatterjee, A., & Sassard, V. (2022). SCENTAR: A high-density nodal array to study the structure and seismogenic behavior of the southern Cascadia forearc. <i>Seismological Research Letters</i> , 94(1), 496–506. https://doi.org/10.1785/0220220251 

Research & Professional Experiences

08/2022 – Present Lafayette, IN, USA	Research and Teaching Assistant, Purdue University
06/2025 – 08/2025	Research Intern, Penn State, Tieyuan Zhu Developed a workflow to process Distributed Acoustic Sensing data in order to apply seismic monitoring of the Susquehanna Shale Hills Critical Zone Observatory.
06/2024	Research Assistant, University of Hawaii, Garrett Apuzen-Ito Selected by Garrett Apuzen-Ito during the Apply-to-Sail program for the recovery of the Marine Imaging of Galápagos Upwelling and Neighboring Archipelago (Marine IGUANA) Ocean Bottom Seismometers (OBS) aboard the R/V Sally Ride from 06/07/2024 (Golfito, Costa Rica) to 07/03/2024 (Puerto Ayora, Galápagos, Ecuador): • Recovered OBS (detection on the surface and tagline). • Analyzed seismic data downloaded directly from the OBS. • Corrected multi-beam data produced by the EM712 and EM124 (MB-system, determines bathymetry).

- Deployed expendable bathythermograph (XBT, determines water properties to create an acoustic wave velocity profile).
- Deployed and recovered magnetometer (determines magnetism in the seafloor).
- Deployed 2 Argo floats (F1578 and F1538, determines water properties to monitor climate change).
- Logged the cruise details.
- Learned to follow safety procedures on a ship.

Volunteer under the supervision of Garrett Apuzen-Ito (UH), Emilie, Hooft (UO), Doug Toomey (UO), and Yang Shen (URI).

National Science Foundation Division of Ocean Sciences Grant #1928197.

07/2023
USA

Volunteer Field Assistant, *Purdue University, Sourav Karmakar*

Helped collect plutonic samples in remote areas of the Northern Cascades for three weeks. Sampled, orientation measurements of fabrics, and drove.

02/2020 – 08/2022
Eugene, OR, USA

Undergraduate Researcher, *University of Oregon, Miles Bodmer*

Testing Mantle Anisotropy Models in the Alaska Subduction Zone through seismic data. NSF Award Number: 1948961.

07/2021 – 08/2021
Ally (43), France

Scientific animator / Tour Guide, *Association Action Ally 2000, Sur le Plateau d'Ally (43)*

Guided visitors in an ancient silver, antimony, and lead mine. Animated a workshop about finding and determining minerals.

04/2020
USA

Volunteer Field Assistant, *University of Oregon, Jonathan Delph*

Helped retrieve a network of seismometers in Northern California (SCENTAR).

Presentations

06/2026

Seismic Constraints on the Structure and Subduction Dynamics along the Alaska Peninsula from Full Wave Ambient Noise Tomography and Shear Wave Splitting,

Workshop on Integrating Seismic Anisotropy, Geodynamics, and Rock Deformation

Authors: **Sassard, V.**, Yang, X., Bodmer, M., Liu, L., Elliott, J. Poster presentation.

06/2026

Shear Wave Splitting Analysis Across the Galápagos Hotspot and Western Galápagos Spreading Center,

Workshop on Integrating Seismic Anisotropy, Geodynamics, and Rock Deformation

Authors: Adeboye, O., Ito, G., **Sassard, V.**, Hooft, E. E. E., Shen, Y., Toomey, D. R., Hufstetler, R. S., Autumn, K., Kongpet, P., & Ruiz, M. C. Poster presentation by Oluwatofunmi Adeboye.

04/2026

Lithosphere Modification and Asthenospheric Upwelling Along Accreted Terrane Boundaries in Interior Alaska: Insights from Full Wave Ambient Noise Tomography,

SZ4D Community Meeting

Authors: **Sassard, V.**, Yang, X., & Ridgway, K. D. Poster presentation.

04/2026

Evidence of Slab Tearing Along a Subducted Oceanic Plate Joint Beneath the Alaska Peninsula, *SZ4D Community Meeting*

Authors: **Sassard, V.**, Yang, X., Liu, L., Elliott, J. Poster presentation.

12/2025

Shear Wave Splitting Across the Galápagos Hotspot and Western Galápagos Spreading Center from the Marine IGUANA Experiment, *AGU Fall Meeting 2025*

Authors: Adeboye, O., Ito, G., **Sassard, V.**, Hooft, E. E. E., Shen, Y., Toomey, D. R., Hufstetler, R. S., Autumn, K., Kongpet, P., & Ruiz, M. C. Poster presentation by Oluwatofunmi Adeboye.

12/2025

Investigating the Upper-Mantle Dynamics of Galapagos Plume-Ridge Interaction with the Marine IGUANA Ocean Bottom Seismometer Array, *AGU Fall Meeting 2025*

Authors: Ito, G., Hooft, E. E. E., Shen, Y., Toomey, D. R., Ruiz, M. C., Hufstetler, R. S., Autumn, K., Adeboye, O., Kongpet, P., **Sassard, V.**, Cheng, Z., & Young, M. Oral presentation by Garrett Ito.

06/2025

Seismic Characteristics of Structural Modifications of the Lithosphere in Alaska due to Shallow Subduction of the Yakutat Slab,

Gordon Research Conference / Gordon Research Seminar

Authors: **Sassard, V.**, Yang, X., & Ridgway, K. D. Poster presentation.

05/2024	A Possible Slab Window Along the Alaska Subduction Zone Imaged by Full Wave Ambient Noise Tomography , <i>SSA Annual Meeting 2024</i> Authors: Sassard, V. , Yang, X., Elliott, J., & Liu, L. Oral presentation.
05/2024	Lithospheric Modification in Northeastern Alaska Interpreted From Full-Wave Ambient Noise Tomography , <i>SSA Annual Meeting 2024</i> Authors: Sassard, V. , Yang, X., Ridgway, K. D., & Flesch, L. Poster presentation.
12/2023	Full-Wave Ambient Noise Tomography Reveals Structural Variations along the Alaska Subduction Zone , <i>AGU Fall Meeting 2023</i> Authors: Sassard, V. , Yang, X., & Elliott J. Oral presentation.
06/2023	Investigating the Segmentation of Subduction Zones Across the Shoreline with Seismic Tomography: Story of the Alaska Subduction Zone , <i>Gordon Research Conference / Gordon Research Seminar</i> Authors: Sassard, V. , & Yang, X. Poster presentation.
04/2023	Subduction Segmentation Revealed by Full-Wave Ambient Noise Tomography of the AACSE Data , <i>Seismological Society of America</i> Authors: Sassard, V. , & Yang, X. Poster presentation.
12/2022	What Controls Lateral Segmentation in Cascadia? Sub-Slab Density Anomalies or Slab Age Variations , <i>AGU Fall Meeting 2022</i> Authors: Sassard, V. , van Dinther, Y., Bodmer, M., Hooft, E. E. E., & Toomey, D. R. Oral presentation.
12/2022	Segmentation of the Aleutian-Alaska Subduction Zone Revealed by Full-Wave Ambient Noise Tomography with the AACSE Data , <i>AGU Fall Meeting 2022</i> Authors: Sassard, V. , & Yang, X. Oral presentation.
12/2021	Evaluating Drivers of Mantle Flow and Sources of Seismic Anisotropy in the Alaskan Subduction Zone: Observations from Offshore/Onshore Shear-Wave Splitting , <i>AGU Fall Meeting 2021</i> Authors: Sassard, V. , & Bodmer, M. Poster presentation. DOI: 10.1002/essoar.10509368.2.
12/2020	Onshore-offshore shear wave splitting observations beneath the Alaska subduction zone: Mantle flow constraints from the Alaska Amphibious Community Seismic Experiment , <i>AGU Fall Meeting 2020</i> Authors: Sassard, V. , & Bodmer, M. Virtual poster presentation.

Academic Services

Reviewer, *Journal of Geophysical Research: Solid Earth, Tectonophysics & Geophysical Research Letters*

Skills

Programming Languages

MATLAB, Python, Generic Mapping Tools, Slurm

Field Geology

Teamwork, moving in remote areas, physical tasks

Languages

English

IELTS 7.5/9 (November 2021)

French

Native

Spanish

Conversational

Honors & Awards

05/2026	Bilsland Dissertation Fellowship , <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i> The Bilsland Dissertation Fellowship is awarded by departments and colleges to PhD students who will enter their final year of their program. Recipients are expected to devote full-time effort to the completion of all requirements necessary to receive their doctoral degree at the conclusion of the year of funding.
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05/2026	William J. Hinze Graduate Award in Exploration Geophysics, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i>
05/2026	Cedric J. Newby Award, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i>
05/2025	Gerald H. Krockover Graduate Fellowship, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i> Awarded for outreach efforts throughout the year.
05/2025	William J. Hinze Graduate Award in Exploration Geophysics, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i> Awarded for pursuing studies that will lead to a career focused on geophysics.
04/2024	Cedric J. Newby Award, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i>
07/2023	Outstanding Student Service Award, <i>Earth, Atmospheric, and Planetary Sciences Department, Purdue University</i>
04/2023	Cedric J. Newby Award, <i>Earth, Atmospheric, and Planetary Sciences, Purdue University</i> "Awarded to students pursuing a degree in solid-earth geosciences leading to professional employment as a geophysicist, who have demonstrated academic proficiency in their chosen field of study, have evidenced a high degree of social, moral, and spiritual character development, and have demonstrated good citizenship and a sense of community responsibility with respect to both Purdue University and their community."
11/2020	Bourse Eole, <i>Réseau Franco-néerlandais (French-Dutch network)</i> Granted for spending my MSc degree in The Netherlands while being French (based on a CV and a cover letter).
06/2020	Earth Sciences Spring Award, <i>Earth Sciences Department, University of Oregon</i> Selected by the Earth Sciences Department of the University of Oregon to receive a Brunton compass awarded from the James Stovall Fellowship Fund.

Teaching Experiences

01/2026 – 05/2026	Computer-Aided Analysis For Geosciences, <i>Purdue University</i> Assisted the students during lab sections and office hours, and graded their work. Gave one lecture on template matching.
01/2026 – 05/2026	Introduction to Earth Sciences, <i>Purdue University</i> Assisted the students during lab sections and office hours, and graded their work. Gave two lectures on earthquakes and seismicity.
08/2025 – 12/2025	Introduction to Earth Sciences, <i>Purdue University</i> Introduced students to various topics, assisted them during two lab sections, and graded their work.
08/2024 – 12/2024	Introduction To Analysis And Computing With Geoscience Data, <i>Purdue University</i> Graded and assisted students in labs and office hours.
01/2024 – 05/2024	Fossil Fuels, Energy, and Society, <i>Purdue University</i> Graded the work of the students.

Short Courses

07/2025	Technical Short Course: Using MsPASS for Data Processing on HPC and Cloud Systems, <i>Earthscope</i> Earthscope-offered technical course taught by Dr. Gary Pavlis on the use of MsPASS and its deployment on high-performance clusters.
03/2024	Gravity and Magnetism for Earth Scientists, <i>Purdue University</i> Learned to interpret gravity and magnetic data in different settings from real data. Learned to use Oasis Montaj, a gravity and magnetism modeling software, on real data from the East African Rift System (Lake Albert). Taught by Mark Longacre from MBL.
06/2020 – 08/2020	IRIS Seismology Skill Building Workshop, <i>Incorporated Research Institutions for Seismology</i> Online summer course. Learned to process seismological data with various tools

Organizations & Leadership Experiences

08/2024 – 05/2026	Graduate Committee, Graduate Student Representative Discussed graduate student matters with the faculty at the department level.
04/2024 – 05/2026	Seminar Committee, Geology and Geophysics Graduate Student Representative Vote, invite, and arrange talks with speakers at both the local and national scale at the Purdue EAPS Department.
08/2024 – 05/2026	Outreach Committee Helped organize outreach activities with the local community. Graduate student-led committee.
08/2024 – 07/2025	Purdue Graduate Student Government, EAPS Senator, Vice-Chair Legislative Review Committee Vote on graduate students matters at the university level. Vice-Chair of the Legislation Review Committee. Volunteered in events with the Career Team.
08/2023 – 06/2024	Diversity, Equity, and Inclusion Committee, Graduate Student Representative Developed an inclusive Field Safety Plan template for the Purdue EAPS Department

Voluntary Experiences

09/2026	Wonders on the Wabash Helped set up canoes with the Tippecanoe County team.
01/2026	Purdue Science Olympiad General Volunteer. Graded and facilitated sessions throughout the day.
01/2026	Science Saturday 3-hour session where we present quick hands-on experiments to visitors on seismology and atmospheric processes.
11/2025	Homecoming Purdue Represented the Computational Seismology and Tectonics group at the EAPS Department booth. Demonstrated how earthquakes can be recorded using an accelerometer.
09/2025	Wonders on the Wabash Helped set up canoes.
04/2025	Spring Fest Helped running the EAPS outreach booth showing crater formation, volcanoes, and water filters.
03/2025	STEAM Camp Organized and co-led two sessions on plate tectonics and atmospheric processes at Imagination with children aged 7 to 11 years old.
09/2024	Homecoming Purdue Represented the Computational Seismology and Tectonics group at the EAPS Department booth. Demonstrated how earthquakes can be recorded using an accelerometer.
09/2024	Wonders on the Wabash Helped set up canoes, and acted as a raft instructor for children with the Tippecanoe County team.
07/2024	SCIENCE Camp Organized and led two sessions at Imagine Station on July 29th and July 30th at Imagination Station in Lafayette. Day 1: Volcanoes, rocks, hazards Day 2: Space, milk galaxy, crater impacts, fossils
04/2024	Spring Fest Helped running the EAPS outreach booth showing crater formation, volcanoes, and water filters.

- 03/2024 **STEAM Camp**
Organized two sessions and led one session about natural hazards (volcanoes, earthquakes, tornadoes) and cloud formation at Imagination Station with children aged 7 to 11 years old.
- 09/2023 **Wonders on the Wabash**
Helped set up canoes, and lunch for children with the Tippecanoe County team.
- 06/2023 **Earth Day at Imagination Station**
Demonstrated how earthquakes are recorded to children during Earth Day along with two other students showing the effects of pollution on animals.
- 09/2022 **Homecoming Purdue**
Represented the Computational Seismology and Tectonics group at the EAPS Department booth. Demonstrated how earthquakes can be recorded using an accelerometer.